

Subnetting Calculations

Dividing 192.168.40.0/24 into two equal subnets

1. Given Network Details

Parameter	Value
Network address	192.168.40.0/24
Original subnet mask	255.255.255.0
Required number of subnets	2
Subnetting method	Borrow 1 host bit

2. Determining the Number of Borrowed Bits

The number of required subnets is 2. The formula used to calculate the number of subnets is:

$$2^n \geq \text{Number of required subnets}$$

For two subnets:

$$2^n \geq 2$$

$$2^1 = 2$$

Therefore, **n = 1** bit is borrowed from the host portion.

3. Calculating the New Subnet Mask

The original network is a /24 network. Borrowing 1 host bit changes the prefix length as follows:

Representation	Value
Original prefix	/24
Borrowed bits	1
New prefix	/25
New subnet mask	255.255.255.128

Binary form of the new subnet mask:

11111111 . 11111111 . 11111111 . 10000000

Decimal form: **255.255.255.128**

4. Subnet Calculation

A /25 subnet contains 128 total addresses. Of these, one address is reserved for the network ID and one for the broadcast address, leaving 126 usable host addresses per subnet.

Subnet	Network ID	Usable Host Range	Broadcast ID
1st Subnet	192.168.40.0/25	192.168.40.1 – 192.168.40.126	192.168.40.127
2nd Subnet	192.168.40.128/25	192.168.40.129 – 192.168.40.254	192.168.40.255

5. Address Block Explanation

First subnet: 192.168.40.0/25

- Network ID: 192.168.40.0
- First usable host: 192.168.40.1
- Last usable host: 192.168.40.126
- Broadcast ID: 192.168.40.127

Second subnet: 192.168.40.128/25

- Network ID: 192.168.40.128
- First usable host: 192.168.40.129
- Last usable host: 192.168.40.254
- Broadcast ID: 192.168.40.255

6. Application in the Packet Tracer Lab

The two subnets were used to separate the ACCOUNTS and DELIVERY departments in the Cisco Packet Tracer topology:

Department	Subnet	Default Gateway
ACCOUNTS	192.168.40.0/25	192.168.40.1
DELIVERY	192.168.40.128/25	192.168.40.129

Conclusion: The /24 network was successfully divided into two equal /25 subnets, providing separate address ranges for the two departments while allowing inter-subnet communication through the router.